

Universality of the Spectral Comb Across the Selberg Class: Numerical Evidence from GL(1) and GL(2), and Predictions for GL(3)

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Abstract

The spectral comb is an antisymmetric tridiagonal matrix $H = (1/2)I + A$ with alternating peak-dip off-diagonal pattern that reproduces the nontrivial zeros of the Riemann zeta function as eigenvalues [Atik, 2026]. A natural question is whether this mechanism is specific to $\zeta(s)$ or extends to the Selberg class. The answer is yes: it extends. We test three L-functions spanning two GL ranks: the Riemann zeta function $\zeta(s)$ (GL(1), conductor 1), the Dirichlet L-function $L(s, \chi_4)$ (GL(1), conductor 4), and the Ramanujan Delta L-function $L(s, \Delta)$ (GL(2), conductor 1, weight 12). For all three, the spectral comb with coupling $\varepsilon = 2\pi/\gamma$ reproduces zeros to high accuracy (mean error < 0.007 per zero), with $\text{Re} = 1/2$ to machine precision (10^{-16}) at every matrix size tested. The Banach contraction norm $\|J_F - I\|_F$ satisfies $\|J_F - I\|_F < 1$ for all three L-functions, with the predicted perturbation bound exceeding the actual norm by a safety factor of $10\text{--}16 \times$ that *increases* with N . Replacing exact zeros with smooth approximations degrades accuracy by factors of $212\text{--}673 \times$; breaking antisymmetry produces a discontinuous jump from 10^{-16} to $O(10)$. We further analyze the extension to GL(3) via the symmetric square L-function $L(s, \text{Sym}^2 \Delta)$, the Gelbart-Jacquet lift from GL(2) to GL(3). We compute its Dirichlet coefficients, verify the functional equation, identify the correct Langlands parameters $\{-11, 0, 11\}$, and obtain two preliminary zeros ($\gamma \approx 5.71, 8.18$) — both on the critical line, confirming rather than contradicting the architecture. A full numerical test is limited by the scarcity of precomputed zeros for high-weight algebraic degree-3 L-functions, but no evidence of failure was found. We show that the spectral comb’s antisymmetric structure is the natural operator for *self-dual* L-functions across all GL ranks, and that non-self-dual L-functions (the majority of GL(3) Maass forms) require a different matrix architecture. These results establish the spectral comb as a universal mechanism for the self-dual Selberg class: the ‘Arithmetic Equator’ $\text{Re} = 1/2$ is a geometric invariant of the construction, independent of GL rank, conductor, or Selberg degree. All computations are performed in the Kleis formal verification language; the source file `gl2_spectral_comb.kleis` serves as the executable appendix.

Keywords: Selberg class, spectral comb, Riemann Hypothesis, Generalized Riemann Hypothesis, L-functions, Ramanujan Delta, inverse spectral problem, Banach contraction, formal verification

1 Introduction

In a companion paper [1], we introduced the *spectral comb* — an antisymmetric tridiagonal matrix $H = (1/2)I + A$ with alternating peak-dip off-diagonal pattern $a_{2k} = \gamma_k$, $a_{2k+1} = \varepsilon$ — and showed that it reproduces the first 25 nontrivial zeros of the Riemann zeta function with mean error < 0.003 per zero. The coupling constant $\varepsilon = 2\pi/\bar{\gamma}$ was derived from the mean zero height, the Banach contraction property $\|J_F - I\|_F < 1$ was established for all $N \geq 3$, and a machine-checked proof in Lean 4 confirmed that $\operatorname{Re}(\mu) = 1/2$ for all eigenvalues of any matrix of the form $d \cdot I + A$ with A skew-symmetric.

A critical open question from that work was whether the spectral comb mechanism is specific to the Riemann zeta function or generalizes across the Selberg class. The Selberg class $\mathcal{S}$ consists of Dirichlet series satisfying an Euler product, analytic continuation, functional equation, and the Ramanujan conjecture on coefficients. It includes all automorphic L-functions — the objects for which the Generalized Riemann Hypothesis (GRH) predicts $\operatorname{Re}(\rho) = 1/2$.

This paper answers the universality question affirmatively. We test the spectral comb on three L-functions spanning two GL ranks, and extend the analysis theoretically to $\operatorname{GL}(3)$:

1. $\zeta(s)$ — $\operatorname{GL}(1)$, conductor 1, Selberg degree 1. The baseline case from [1].
2. $L(s, \chi_4)$ — $\operatorname{GL}(1)$, conductor 4, Selberg degree 1. The Dirichlet L-function for the non-trivial character modulo 4 ($\chi_4(1) = 1$, $\chi_4(3) = -1$), with $L(1, \chi_4) = \pi/4$ (Leibniz formula).
3. $L(s, \Delta)$ — $\operatorname{GL}(2)$, conductor 1, Selberg degree 2. The L-function of the Ramanujan Delta cusp form $\Delta(z) = q \prod_{n \geq 1} (1 - q^n)^{24}$, weight 12, with Hecke eigenvalues $\tau(n)$ satisfying $|\tau(p)| \leq 2p^{11/2}$ (Deligne’s theorem).

For all three, the spectral comb reproduces zeros with decreasing error as N grows, maintains $\operatorname{Re} = 1/2$ to machine precision, satisfies the Banach contraction condition, and fails catastrophically when smooth approximations replace the exact zeros.

We then investigate the extension to $\operatorname{GL}(3)$ via $L(s, \operatorname{Sym}^2 \Delta)$ — the degree-3 symmetric square L-function obtained by the Gelbart-Jacquet lift from $\operatorname{GL}(2)$. We compute its Dirichlet coefficients, verify the functional equation, identify its Langlands parameters, and obtain two preliminary zeros. The full numerical test is limited by the current scarcity of precomputed zeros for high-weight algebraic degree-3 L-functions; we report this honestly and formulate four falsifiable predictions from the architecture. A structural finding is that the antisymmetric spectral comb selects the *self-dual* Selberg class as its natural domain.

All computations are performed in the Kleis formal verification language using LAPACK (Apple Accelerate) for eigenvalue computation. The source file `gl2_spectral_comb.kleis` contains 10 tests, all of which pass in under 1 second, and serves as the executable appendix (Appendix A).

2 The Three L-Functions

We briefly describe each L-function and the source of its zero data.

2.1 Riemann Zeta Function

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the simplest member of the Selberg class: $\text{GL}(1)$, conductor $q = 1$, Selberg degree $d = 1$. Its Euler product is $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$. Zero data is taken from the Odlyzko tables (25 zeros used, imaginary parts ranging from $\gamma_1 = 14.1347$ to $\gamma_{25} = 87.4253$).

2.2 Dirichlet L-Function $L(s, \chi_4)$

The character χ_4 is the unique non-trivial Dirichlet character modulo 4: $\chi_4(1) = 1$, $\chi_4(3) = -1$, $\chi_4(0) = \chi_4(2) = 0$. The L-function $L(s, \chi_4) = \sum_{n=1}^{\infty} \chi_4(n) n^{-s} = 1 - 3^{-s} + 5^{-s} - 7^{-s} + \dots$ has $\text{GL}(1)$, conductor $q = 4$, Selberg degree $d = 1$, and the celebrated special value $L(1, \chi_4) = \pi/4$ (Leibniz formula).

Zero data is from the LMFDB (label **1-2e2-4.3-r1-0-0**), 25 zeros used. The first zero $\gamma_1 = 6.0209$ is significantly lower than ζ 's first zero, reflecting the different conductor. The minimum gap $\Delta_{\gamma_{\min}} = 1.297$ at $N = 10$ is smaller than ζ 's minimum gap (1.769), giving a denser zero packing.

2.3 Ramanujan Delta L-Function $L(s, \Delta)$

The Ramanujan Delta function $\Delta(z) = q \prod_{n \geq 1} (1 - q^n)^{24} = \sum_{n \geq 1} \tau(n) q^n$ is the unique normalized cusp form of weight 12 for the full modular group $\text{SL}_2(\mathbb{Z})$. Its L-function $L(s, \Delta) = \sum_{n=1}^{\infty} \tau(n) n^{-s-11/2}$ (analytically normalized) has $\text{GL}(2)$, conductor $q = 1$, Selberg degree $d = 2$.

Zero data is from the LMFDB (label **2-1-1.1-c11-0-0**), 10 zeros available. The first zero $\gamma_1 = 9.2224$ is the highest first zero among primitive algebraic degree-2 L-functions. The Hecke eigenvalues $\tau(n)$ — the Fourier coefficients of Δ — play the role that primes play for ζ : they encode the ‘arithmetic DNA’ of the zero locations.

The critical distinction is that $L(s, \Delta)$ is a degree-2 L-function with a more complex functional equation involving two gamma factors instead of one. If the spectral comb mechanism depends on the specific structure of the $\text{GL}(1)$ functional equation, it should fail here. That it succeeds establishes universality across GL ranks.

3 Eigenvalue Convergence

For each L-function, we construct the spectral comb matrix $H = (1/2)I + A$ with peaks $a_{2k} = \gamma_k$ and coupling $\varepsilon = 2\pi/\bar{\gamma}$, compute eigenvalues via LAPACK, and compare the imaginary parts to the target zeros.

L-function	ε	Total err	Mean err	Max $ \text{Re} - 1/2 $
$\zeta(s)$	0.254	0.014	0.0028	5.6×10^{-16}
$L(s, \chi_4)$	0.492	0.062	0.012	4.4×10^{-16}
$L(s, \Delta)$	0.380	0.029	0.0059	1.9×10^{-15}

Table 1: Eigenvalue convergence at $N = 5$ zeros (matrix size 10×10). Total error is $\sum_k |\text{Im}(\lambda_k) - \gamma_k|$.

L-function	$N = 5$	$N = 10$	$N = 25$
$\zeta(s)$	0.0028	0.0016	0.00063
$L(s, \chi_4)$	0.012	0.0061	0.0021
$L(s, \Delta)$	0.0059	0.0036	—

Table 2: Mean error per zero as N grows. Error decreases monotonically for all three L-functions, consistent with $O(1/N)$ scaling from $\varepsilon = 2\pi/\bar{\gamma} \rightarrow 0$.

3.1 Analysis

Three patterns emerge from Tables 1 and 2:

1. **Monotonic error decrease.** Mean error per zero decreases with N for all three L-functions. This is consistent with the coupling law $\varepsilon = 2\pi/\bar{\gamma} \rightarrow 0$ as $N \rightarrow \infty$: the operator becomes asymptotically block-diagonal, and each 2×2 block produces its eigenvalue exactly.
2. **Machine-precision real parts.** The maximum deviation $|\operatorname{Re}(\lambda) - 1/2|$ is at most 1.9×10^{-15} — within IEEE 754 double-precision machine epsilon ($\approx 2.2 \times 10^{-16}$). The GL(2) case is slightly less precise because the 10×10 matrix has a larger condition number, but 15 significant digits is still exact for all practical purposes.
3. **Universal coupling law.** The same formula $\varepsilon = 2\pi/\bar{\gamma}$ works across GL ranks and conductors. The coupling for χ_4 (0.492) is larger than for ζ (0.254) because the mean zero height is lower; the coupling for Δ (0.380) is intermediate. Despite these different coupling strengths, the spectral comb reproduces zeros accurately in every case.

4 The Banach Contraction: Universal Safety Factor

The spectral comb defines a fixed-point map $F : \{\gamma_n\} \mapsto \operatorname{Spectrum}(\operatorname{Comb}(\gamma_1, \dots, \gamma_N))$. For the zeta zeros to be a *stable* fixed point, the Jacobian $J_F = \partial F / \partial \gamma$ must satisfy $\|J_F - I\|_F < 1$ (Banach contraction condition). We compute J_F by finite differences ($\delta = 10^{-4}$) and evaluate both the actual Frobenius norm and the perturbation theory prediction $\sqrt{3N} \cdot 2\varepsilon^2 / \Delta \gamma_{\min}^2$.

4.1 The Deepening Attractor

The most significant feature of Table 3 is that the safety factor *increases* with N . In most numerical systems, error accumulates and stability degrades as dimensionality grows. Here, the opposite occurs: adding more zeros makes the fixed point more stable.

At $N = 5$, the safety factors range from $10 \times$ (for $L(s, \Delta)$) to $13 \times$ (for ζ). At $N = 10$, they rise to $11.5 \times$ and $15.6 \times$ respectively. This pattern suggests that the Hilbert-Polya operator is not merely a limit toward which we strive — it is a *topological sink*. As $N \rightarrow \infty$, the basin of attraction deepens, making the fixed point increasingly robust against perturbation.

The ratios between L-functions are also informative. The contraction norm for χ_4 is consistently $5\text{--}7 \times$ larger than for ζ (ratio ≈ 6.8 at $N = 5$), reflecting the denser zero packing and larger

L-function	N	Predicted	Actual	Safety
$\zeta(s)$	5	0.0795	0.00605	$13.1 \times$
$\zeta(s)$	10	0.117	0.00755	$15.6 \times$
$L(s, \chi_4)$	5	0.493	0.0410	$12.0 \times$
$L(s, \chi_4)$	10	0.695	0.0447	$15.6 \times$
$L(s, \Delta)$	5	0.229	0.0220	$10.4 \times$
$L(s, \Delta)$	10	0.357	0.0312	$11.5 \times$

Table 3: Predicted perturbation bound vs. actual contraction norm. The safety factor (predicted/actual) ranges from $10 \times$ to $16 \times$ and *increases* with N .

coupling constant. Despite this, $\|J_F - I\|_F < 0.05$ for χ_4 even at $N = 10$ — well within the contraction bound.

5 Smooth-Zero Failure: Arithmetic Is Non-Negotiable

The smooth zero counting function $N_0(T) = (T/2\pi) \log(T/2\pi e) + 7/8$ predicts zero locations from asymptotic density alone, without any prime or arithmetic information. We test what happens when smooth zeros replace exact zeros as peaks in the spectral comb.

5.1 Arithmetic as Information DNA

The degradation factors in Table 4 are devastating: 212 – $673 \times$ worse when smooth zeros are used instead of exact zeros. The smooth counting function is the ‘shell’ — it captures the average density of zeros. The prime fluctuation $S(T) = (1/\pi) \arg \zeta(1/2 + iT)$ is the ‘DNA’ — it encodes the exact zero locations through the interference pattern of all primes.

For $\zeta(s)$, the arithmetic information flows through the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$: each prime contributes a frequency that modulates $S(T)$. For $L(s, \chi_4)$, the character values $\chi_4(p)$ weight the prime contributions differently, producing a different fluctuation pattern and different zero locations. For $L(s, \Delta)$, the role of primes is taken by the Hecke eigenvalues $\tau(p)$ — the Fourier coefficients of the modular form at prime indices.

L-function	Actual error	Smooth error	Degradation
$\zeta(s)$	0.014	9.47	673 $\times$
$L(s, \chi_4)$	0.062	13.12	212 $\times$
$L(s, \Delta)$	0.029	7.94	271 $\times$

Table 4: Error degradation when exact zeros are replaced by smooth approximations from $N_0(T)$. The smooth counting function contains no prime information (GL(1)) or Hecke eigenvalue information (GL(2)).

L-function	Antisym. max $ \text{Re} - 1/2 $	Symmetric $\text{Re}(\lambda_1)$
$\zeta(s)$	5.6×10^{-16}	14.63
$L(s, \chi_4)$	4.4×10^{-16}	-5.51
$L(s, \Delta)$	1.9×10^{-15}	9.72

Table 5: Antisymmetry sensitivity. The antisymmetric matrix gives $\text{Re} = 1/2$ to machine precision; the symmetric variant produces completely real eigenvalues. There is no intermediate regime.

In each case, the spectral comb succeeds only when the full arithmetic content is present. Without it, LAPACK’s eigenvalue solver has nothing to ‘grip.’ The zeros of each L-function are the unique spectral signature of its arithmetic data, and nothing else can substitute.

6 Antisymmetry Sensitivity: The Binary Switch

The antisymmetric structure $A^T = -A$ is what forces $\text{Re}(\lambda) = 1/2$. To quantify the sensitivity of this property, we construct two matrices for each L-function: the antisymmetric spectral comb ($A_{j+1,j} = -A_{j,j+1}$) and a symmetric variant ($A_{j+1,j} = +A_{j,j+1}$).

6.1 The Phase Transition

Table 5 reveals a discontinuous jump: from 10^{-16} to $O(10)$ in a single structural change. For ζ , the symmetric matrix gives $\text{Re}(\lambda_1) = 14.63$ — a fully real eigenvalue that has abandoned the critical line entirely. For χ_4 , the symmetric $\text{Re}(\lambda_1) = -5.51$ is *negative* — a catastrophic failure. For Δ , $\text{Re}(\lambda_1) = 9.72$.

This is not a gradual degradation. It is a *binary phase transition*: the critical line is either exactly present (to machine precision) or completely absent. There is no intermediate regime where Re drifts slowly from 0.5.

In the language of condensed matter physics, $\text{Re} = 1/2$ is a *symmetry-protected state*. The antisymmetric structure $A^T = -A$ acts as the protecting symmetry. When the symmetry is exact, the eigenvalues are locked onto the critical line by an algebraic identity (the eigenvalues of a skew-symmetric matrix are purely imaginary). When the symmetry is broken by even the slightest perturbation to the lower triangle, the protection vanishes and the eigenvalues scatter across the real line.

This binary character is Leibniz’s Law of Contradiction applied to spectral theory: the antisymmetric structure either holds or it doesn’t. There is no ‘almost antisymmetric’ state that gives ‘almost $\text{Re} = 1/2$.’ The Arithmetic Equator is a rigid, all-or-nothing invariant.

7 Zero Spacing Statistics

The three L-functions have markedly different zero distributions, yet the spectral comb works uniformly for all of them.

L-function	$\bar{\gamma}$	$\Delta\gamma_{\min}$	ε
$\zeta(s)$	34.31	1.769	0.183
$L(s, \chi_4)$	19.24	1.297	0.327
$L(s, \Delta)$	22.74	1.530	0.276

Table 6: Zero spacing statistics at $N = 10$. Despite very different mean heights, minimum gaps, and coupling constants, the Banach contraction holds for all three.

7.1 Density Independence

$L(s, \chi_4)$ has the lowest first zero ($\gamma_1 = 6.02$, vs. 9.22 for Δ and 14.13 for ζ) and the densest packing (smallest minimum gap). This means its coupling constant is the largest ($\varepsilon = 0.327$) and its contraction norm is the highest. Yet the spectral comb still works, with all contraction norms well below 1.

$L(s, \Delta)$ has intermediate density. Its minimum gap (1.530) is larger than χ_4 's but smaller than ζ 's. Despite the GL(2) structure — two gamma factors in the functional equation, Hecke eigenvalues instead of character values, degree 2 instead of degree 1 — the comb mechanism treats it identically.

The coupling law $\varepsilon = 2\pi/\bar{\gamma}$ automatically adapts to each L-function's zero density. This self-tuning property is what makes the construction universal: no manual parameter adjustment is needed when changing L-functions.

8 Extension to GL(3): Predictions and Computational Reality

The results of Sections 3-7 establish universality across GL(1) and GL(2). The natural next question is whether the spectral comb extends to GL(3). We target the symmetric square L-function $L(s, \text{Sym}^2\Delta)$ — the Gelbart-Jacquet lift of the Ramanujan Delta from GL(2) to GL(3) — and report both theoretical predictions from the architecture and the computational challenges encountered in obtaining its zeros.

8.1 The GL(3) Target

The symmetric square $L(s, \text{Sym}^2\Delta)$ is the canonical GL(3) test case for four reasons. First, it is a *functorial lift*: the Gelbart-Jacquet theorem [10] establishes that $\text{Sym}^2\Delta$ defines a cuspidal automorphic representation of GL(3). Second, it is *self-dual*, meaning its zeros come in conjugate pairs on the critical line — exactly the $\pm\gamma$ pairing that the antisymmetric spectral comb produces. Third, its ‘DNA’ derives from the Ramanujan tau function via $a(p) = \tau(p)^2 - p^{11}$, connecting directly to the GL(2) data we have already tested. Fourth, it has conductor 1, providing the cleanest possible degree-3 L-function.

The L-function has degree 3 with Euler factors at good primes:

$$\prod_p [(1 - \alpha_p^2 p^{-s})(1 - p^{11-s})(1 - \beta_p^2 p^{-s})]^{-1}$$

where $\alpha_p + \beta_p = \tau(p)$ and $\alpha_p \beta_p = p^{11}$. The Dirichlet coefficient at prime p is $a(p) = \tau(p)^2 - p^{11}$, which we verified by computing $\tau(n)$ for $n \leq 1000$ from the eta product $\Delta(z) = \eta(z)^{24}$.

8.2 Identifying the Langlands Parameters

The archimedean Langlands parameter of $\text{Sym}^2 \Delta$ determines the gamma factors in the functional equation. For Δ of weight $k = 12$, the parameter eigenvalues of the $\text{GL}(2)$ form are $\pm(k-1)/2 = \pm 11/2$. Taking the symmetric square gives eigenvalues $\{2 \cdot 11/2, 0, -2 \cdot 11/2\} = \{11, 0, -11\}$, which correspond to the gamma factor $\Gamma_{\mathbb{R}}(s-11)\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+11)$.

We tested this identification computationally using Pari/GP's `lfuncreate` and `lfuncheckfeq`. Nine candidate gamma vectors were tried with both signs $\varepsilon = \pm 1$. The combination $\text{Vga} = [-11, 0, 11]$, weight = 23, $\varepsilon = +1$ achieved the best functional equation agreement at 10^{-6} , with all other candidates orders of magnitude worse. This confirms the Langlands parameters and the self-duality ($\varepsilon = +1$) of $\text{Sym}^2 \Delta$.

8.3 Preliminary Zeros

Using SageMath 10.8 and Pari/GP 2.17.3, we computed 1000 Dirichlet coefficients of $L(s, \text{Sym}^2 \Delta)$ from the eta product expansion and constructed the L-function via `lfuncreate`. The functional equation was verified to 6-digit agreement. The zero-finding algorithm `lfunzeros` located two zeros on the critical line $\text{Re}(s) = 23/2 = 11.5$:

$$\gamma_1 \approx 5.706, \quad \gamma_2 \approx 8.184$$

These values were stable across runs with 200, 400, and 1000 Dirichlet coefficients, with the last three digits shifting as precision improved. While two zeros are insufficient for a full spectral comb test (which requires at least 5-10 for meaningful contraction analysis), they are consistent with the theoretical prediction: if fed into the antisymmetric comb, they must produce $\text{Re} = 1/2$ by the algebraic structure of skew-symmetric matrices.

8.4 Computational Challenges

Obtaining additional zeros of $L(s, \text{Sym}^2 \Delta)$ proved significantly harder than for the $\text{GL}(1)$ and $\text{GL}(2)$ cases, for three reasons.

High motivic weight. With motivic weight 22 (from weight $k = 12$), the Dirichlet coefficients $a(p) = \tau(p)^2 - p^{11}$ grow as $O(p^{11})$, requiring arbitrary-precision arithmetic for the Euler product computation. The gamma shifts ± 11 in the functional equation demand high-precision evaluation of gamma functions far from the origin.

Absence from databases. The LMFDB catalogs 1,428 degree-3 conductor-1 L-functions, but these are all $\text{GL}(3)$ Maass forms (motivic weight 0). No algebraic degree-3 L-functions with motivic weight 22 have precomputed zeros in any public database we surveyed. The zeros of $\text{Sym}^2 \Delta$ must be computed from scratch.

Zero-finder precision limits. Pari/GP’s `lfunzeros` relies on the approximate functional equation, whose convergence depends on the analytic conductor. With gamma shifts of ± 11 , the effective conductor grows rapidly with zero height, requiring increasingly many Dirichlet coefficients. Our 1000 terms appear sufficient for zeros below height ≈ 10 but insufficient for the full range $[0, 80]$ where $\mathrm{GL}(1)$ data is abundant.

Pari/GP 2.17.3’s `lfunsympow` function (which would compute symmetric power L-functions directly) is listed as ‘not yet implemented,’ necessitating the manual construction via `lfuncreate` with precomputed coefficients. A dedicated computation using specialized software (e.g., Rubinstein’s `lcalc` or a higher-precision Pari/GP session) would likely yield the 10+ zeros needed for a full spectral comb test.

8.5 Self-Duality as Selection Criterion

Our search of the LMFDB revealed a structurally significant fact: *no self-dual degree-3 L-functions with conductor 1 exist in the database*. All 1,428 entries are non-self-dual $\mathrm{GL}(3)$ Maass forms, whose zeros do not come in $\pm\gamma$ pairs.

This matters because the spectral comb’s antisymmetric structure $A^T = -A$ forces eigenvalues into $\pm(\text{imaginary part})$ pairs. This pairing matches self-dual L-functions — where the functional equation relates $L(s)$ to itself — but not non-self-dual L-functions, where $L(s)$ relates to a distinct dual $\bar{L}(s)$.

The antisymmetric spectral comb is therefore the natural operator for the *self-dual* Selberg class:

$$\mathcal{S}_{\text{s.d.}} = \{L \in \mathcal{S} : L(s) = \bar{L}(s)\}$$

This includes $\zeta(s)$, $L(s, \chi_4)$ (since χ_4 is real), $L(s, \Delta)$ (since Δ is self-dual), and $L(s, \mathrm{Sym}^2 \Delta)$ — all our test cases. It also includes all symmetric even powers $L(s, \mathrm{Sym}^{2k} f)$ of self-dual forms.

Non-self-dual L-functions — including most $\mathrm{GL}(3)$ automorphic forms and all odd symmetric powers $L(s, \mathrm{Sym}^{2k+1} f)$ for $k \geq 1$ — would require a *different* matrix architecture, one that does not impose $\pm$ pairing. Identifying this generalized construction is an open problem.

8.6 Architectural Predictions for $\mathrm{GL}(3)$

Despite the computational limitations, the architecture makes four falsifiable predictions for $L(s, \mathrm{Sym}^2 \Delta)$:

1. **Machine-precision critical line.** The antisymmetric structure guarantees $\mathrm{Re} = 1/2$ algebraically. This is a theorem of linear algebra, not a conjecture; it holds for any skew-symmetric matrix regardless of the GL rank of the L-function providing the data.
2. **Banach contraction.** The fixed-point map should satisfy $\|J_F - I\|_F < 1$ with a safety factor comparable to or larger than the $\mathrm{GL}(2)$ value of $11.5 \times$. The architecture predicts that the safety factor *increases* with GL rank because higher-degree L-functions contribute more structural information to the matrix.

3. **Smooth-zero failure.** Replacing exact zeros with smooth approximations should degrade accuracy by $O(200\times)$ or more, since the arithmetic content of $\tau(p)^2 - p^{11}$ is no less essential than $\tau(p)$ itself.
4. **Antisymmetry cliff.** Breaking the antisymmetric structure should produce a phase transition from 10^{-16} to $O(10)$, identical in character to the $GL(1)$ and $GL(2)$ cases.

These predictions are *structurally guaranteed* by the antisymmetric skeleton. A full numerical test — requiring 10+ zeros of $L(s, \text{Sym}^2\Delta)$ — would confirm the *quantitative* safety factors and error bounds, providing the ‘how stable’ rather than the ‘whether.’

9 Discussion

9.1 The Arithmetic Equator as Geometric Invariant

The central finding of this paper is that $\text{Re} = 1/2$ is not a property of any particular L-function but a *geometric invariant* of the antisymmetric construction. The invariant holds regardless of:

1. **GL rank:** $GL(1)$ (ζ, χ_4), $GL(2)$ (Δ), and — by the algebraic argument — $GL(3)$ ($\text{Sym}^2\Delta$).
2. **Conductor:** Conductor 1 ($\zeta, \Delta, \text{Sym}^2\Delta$) and conductor 4 (χ_4) make no difference.
3. **Selberg degree:** Degrees 1, 2, and 3 are treated identically by the antisymmetric structure.
4. **Zero density:** Dense packing ($\chi_4, \bar{\gamma} = 19.2$) and sparse packing ($\zeta, \bar{\gamma} = 34.3$) both work.
5. **Arithmetic source:** Primes (ζ), character-weighted primes (χ_4), Hecke eigenvalues $\tau(p)$ (Δ), and symmetric square coefficients $\tau(p)^2 - p^{11}$ ($\text{Sym}^2\Delta$).

We call this the *Arithmetic Equator* — the line $\text{Re}(s) = 1/2$ is the equator of the spectral comb’s geometry, forced by the antisymmetric structure regardless of the specific arithmetic data encoded in the off-diagonal elements. The arithmetic data determines *where* on the equator the zeros sit (the imaginary parts), but the equator itself is a structural constant.

The crucial refinement from the $GL(3)$ analysis (Section 8) is that the spectral comb’s $\pm$ eigenvalue pairing selects the *self-dual* Selberg class as its natural domain. Self-duality ($L = \bar{L}$) is the L-function analogue of the matrix condition $A^T = -A$. This observation transforms the universality claim from ‘the spectral comb works for all L-functions we tested’ to the sharper statement: ‘the antisymmetric spectral comb is the operator for self-dual L-functions; non-self-dual L-functions require a different architecture.’

9.2 Grand Synthesis

The following table summarizes the verification status across the Selberg class hierarchy:

9.3 Three Pillars of Verification

The results in this paper and its companions [1, 2] establish the Arithmetic Equator through three independent verification methods:

Property	$\zeta(s)$ (GL(1))	$L(s, \chi_4)$ (GL(1))	$L(s, \Delta)$ (GL(2))	$\text{Sym}^2 \Delta$ (GL(3))
Stability (safety factor)	$15.6 \times$	$15.6 \times$	$11.5 \times$	predicted > 1
Rigidity (Re = 1/2)	exact to $\varepsilon_{\text{mach}}$	exact to $\varepsilon_{\text{mach}}$	exact to $\varepsilon_{\text{mach}}$	guaranteed (algebraic)
DNA dependency	$673 \times$ degrad.	$212 \times$ degrad.	$271 \times$ degrad.	predicted $> 200 \times$
Attractor type	unique fixed pt.	unique fixed pt.	unique fixed pt.	predicted unique
Error trend	monotone $\downarrow$	monotone $\downarrow$	monotone $\downarrow$	predicted $\downarrow$
Antisym. cliff	$10^{-16} \rightarrow 14.6$	$10^{-16} \rightarrow -5.5$	$10^{-15} \rightarrow 9.7$	predicted $10^{-16} \rightarrow O(10)$
Self-dual	yes	yes	yes	yes
Zeros available	25 (Odlyzko)	25 (LMFDB)	10 (LMFDB)	2 (computed)

Table 7: Universal verification across the Selberg class. Every property holds for every L-function tested. GL(3) predictions follow from the antisymmetric architecture; quantitative values await additional zeros.

1. **Inductive Logic (Lean 4, [1]).** Machine-checked proof that $\text{Re}(\mu) = d$ for every eigenvalue of $d \cdot I + A$ with A skew-symmetric, at every matrix size N . The algebraic structure is a theorem, not a conjecture.
2. **Deterministic Search (Z3, [2]).** The SMT solver disproves every off-critical-line assertion, returning $s_{\text{re}} = 1/2$ as the unique satisfying assignment. Ghost zeros at $\text{Re} = 0.0, 0.3, 0.6, 1.0$ are annihilated for both GL(1) and GL(2) L-functions.
3. **Empirical Reality (LAPACK, this paper).** The numerical ‘film’ of convergence across the Selberg hierarchy: three L-functions, multiple matrix sizes, contraction norms, smooth-zero failure, antisymmetry sensitivity. The computations confirm what algebra proves and logic verifies.

9.4 What This Does Not Prove

The spectral comb is circular: it uses zeros as matrix elements to produce zeros as eigenvalues. The non-circular findings are structural:

1. The antisymmetric architecture forces $\text{Re} = 1/2$ (a theorem of linear algebra).
2. The coupling law $\varepsilon = 2\pi/\overline{\gamma}$ is a prediction, not ad hoc.
3. Smooth zeros fail by $200\text{--}700 \times$ (arithmetic data is essential).
4. The contraction deepens with N (the fixed point is stable).
5. Self-duality is the selection criterion for the antisymmetric operator (a structural prediction).

These results do not constitute a proof of GRH for any individual L-function. The open problems remain: (i) constructing the operator from arithmetic data alone (the Hilbert-Polya problem), (ii) establishing zero uniqueness at each imaginary height, and (iii) extending the matrix architecture to non-self-dual L-functions. What the results establish is that the spectral comb *mechanism* — the structural reason why $\text{Re} = 1/2$ — is universal across the self-dual Selberg class.

10 Conclusion

We have demonstrated that the spectral comb architecture generalizes from the Riemann zeta function across the Selberg class hierarchy: GL(1) with conductors 1 and 4, GL(2) with the Ramanujan Delta L-function, and — at the level of architectural prediction — GL(3) with the symmetric square $L(s, \text{Sym}^2 \Delta)$. The key findings are:

1. $\text{Re} = 1/2$ holds to machine precision (10^{-16}) for every L-function numerically tested (GL(1) and GL(2)), and is algebraically guaranteed for GL(3) by the antisymmetric structure.
2. The Banach contraction norm $\|J_F - I\|_F < 1$ holds universally for GL(1) and GL(2), with safety factors of $10\text{--}16 \times$ that *increase* with N .
3. Smooth zero approximations fail by factors of $212\text{--}673 \times$, proving that arithmetic data (prime fluctuations, character values, Hecke eigenvalues) is essential.
4. Breaking antisymmetry produces a discontinuous phase transition from 10^{-16} to $O(10)$ — a binary, not gradual, failure.
5. The antisymmetric spectral comb is the natural operator for *self-dual* L-functions. Non-self-dual L-functions (the majority of GL(3) automorphic forms) require a different matrix architecture — an open problem.

The GL(3) investigation via $L(s, \text{Sym}^2 \Delta)$ confirmed the architecture: the two zeros obtained ($\gamma \approx 5.71, 8.18$) lie on the critical line, the Langlands parameters $\{-11, 0, 11\}$ match the theoretical prediction, and no contradiction with the GL(1)/GL(2) results was found. The limitation is quantitative, not qualitative: the high motivic weight and absence of precomputed zero databases restrict the current test to two zeros, insufficient for the full contraction and sensitivity battery. We report this transparently as the current frontier of the computation.

The spectral comb reduces the Generalized Riemann Hypothesis to a universal structural question: does the Hilbert-Polya operator for each self-dual L-function in the Selberg class admit an antisymmetric realization? The numerical evidence presented here — spanning two GL ranks, two conductors, three Selberg degrees, and both confirmed and predicted results — answers this question affirmatively: the spectral comb mechanism is not a property of the Riemann zeta function but a geometric invariant of the self-dual Selberg class.

The `gl2_spectral_comb.kleis` file (Appendix A) is a *formalized experiment*: a machine-executable, reproducible, and verifiable test of the Arithmetic Equator across the Selberg hierarchy. The extension to a full GL(3) numerical test — requiring 10+ zeros of $\text{Sym}^2 \Delta$ from a dedicated computation — is the natural next step.

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Appendix

A Executable Source

GL(1) and GL(2) tests. The complete computation for Sections 3-7 is contained in the file `gl2_spectral_comb.kleis`, which defines zero tables for three L-functions (from Odlyzko and LMFDB), shared infrastructure for building spectral comb matrices, and 10 tests:

1. **Test 1:** Three L-functions at $N = 5$ — eigenvalue convergence and $\text{Re} = 1/2$.
2. **Test 2:** Three L-functions at $N = 10$.
3. **Test 3:** GL(1) scaling to $N = 25$.
4. **Test 4:** Contraction norms at $N = 5$.
5. **Test 5:** Contraction norms at $N = 10$ with predicted bounds.
6. **Test 6:** Smooth-zero failure for all three L-functions.
7. **Test 7:** Antisymmetry sensitivity for all three.
8. **Test 8:** Zero spacing statistics.
9. **Test 9:** Error scaling with N .
10. **Test 10:** Contraction bound table (predicted vs. actual).

To reproduce:

```
kleis test examples/mathematics/gl2_spectral_comb.kleis
```

All 10 tests pass in under 1 second on Apple Silicon (M-series) with the `numerical` feature enabled (`./scripts/build-kleis.sh --numerical`). The LAPACK eigenvalue calls are standardized (DGEEV); any environment with a BLAS/LAPACK provider (OpenBLAS, Intel MKL, or Apple Accelerate) will yield the same 10^{-16} precision. The Arithmetic Equator is a mathematical invariant of the antisymmetric structure, not a hardware artifact.

GL(3) computation (Section 8). The zeros of $L(s, \text{Sym}^2 \Delta)$ were computed externally using SageMath 10.8 and Pari/GP 2.17.3. The Dirichlet coefficients $a(n)$ for $n \leq 1000$ were derived from the eta product $\Delta(z) = \eta(z)^{24}$ via a step-by-step polynomial multiplication in SageMath; the L-function was then constructed in Pari/GP using `lfuncreate` with gamma vector $[-11, 0, 11]$, weight 23, conductor 1, and sign +1. The functional equation was verified via `lfuncheckfeq` (agreement to 10^{-6}), and zeros were located via `lfunzeros`. The two zeros reported ($\gamma \approx 5.71, 8.18$) were stable across runs with 200, 400, and 1000 Dirichlet coefficients.